

Appendix A – Risk Rules (17 Cases)

This appendix lists the 17 risk cases that the AI service uses to classify the behaviour of each tracked pedestrian or wheeled vehicle (see section 6.4). The rules are implemented in [backend/ai-service/risk_rules.py](https://github.com/ai-service/risk_rules.py).

A.1 Units and Terms

- **h (body height)** – the height of the person's bounding box. Distances are measured in h, so the rules do not depend on how far the camera is.
- **h/s** – body heights per second. Normal walking is about 0.8–1.5 h/s; running is 2.0 h/s or more.
- **The edge** – the curb line of the crosswalk zone. A positive distance means the person is still on the sidewalk.
- **Not slowing** – the approach speed in the last 0.5 seconds has not dropped.
- **Holding a phone** – a phone was detected near the person's chest in at least 50% of the recent frames.

A.2 The Rules

The table is ordered by priority. When several rules match the same person, the first matching rule in this order is used.

ID	Level	Name	Condition
H6	High	Child running toward the road	A child approaching the edge at 2.0 h/s or faster.
H1	High	Bursting toward the road	Approaching the edge at 2.0 h/s or faster (running).
H5	High	Entering the road without stopping	Approaching at 1.5 h/s or faster, not slowing, and less than 1.2 h from the edge.
H3	High	Distracted and not stopping at the edge	Approaching while holding a phone, not slowing, and less than 0.5 h from the edge.
H4	High	Sudden appearance near the edge	A person first seen less than 0.6 seconds ago, already near the edge and moving toward it.
HW	High	Wheeled vehicle approaching fast	A bicycle or motorcycle within 2.5 h of the edge, approaching at 2.5 h/s or faster, or not slowing.
H2	Medium	Child approaching the road	A child walking toward the edge, less than 1.2 h away or expected to reach it within 2 seconds.
M2	Medium	Distracted approach (phone)	Approaching while holding a phone, less than 1.5 h from the edge.
M5	Medium	Brisk approach without slowing	Approaching at 1.5–2.0 h/s, not slowing, and still 0.5 h or more from the edge.
MW	Medium	Wheeled vehicle slowing, but not enough	A bicycle or motorcycle within 2.5 h of the edge that is slowing down but still approaching at 0.5 h/s or faster.
M4	Low	Presence in the crossing zone	Stepping off the curb, or standing still past the curb line.
L4	Low	Quiet waiting at the edge	Standing still less than 0.5 h from the edge.
M1	Low	Normal approach to the crossing	Walking toward the edge at a normal pace, less than 1.2 h away or expected to reach it within 2 seconds.
L1	Low	Moving parallel to the road	Moving along the road, not toward it.

ID	Level	Name	Condition
L3	Low	Moving away	Moving away from the crossing.
L2	Low	Standing far from the edge	Standing still 0.5 h or more from the edge.
M3	Low	Group of pedestrians approaching	At least 3 people approaching together, the closest of them less than 1.5 h from the edge.

A.3 Notes

- **Action by level:** Low alerts are only logged. Medium and High alerts are also flagged to activate the LED warning (`ledTriggered = true`).
- **Rule IDs:** the letter in the ID is a historical name and does not always match the level. M1, M3 and M4 are Low, and H2 is Medium. The level column is the one that counts.
- **Children:** a child is identified by body height (less than 0.9 of the adult reference height at the same position in the image). This requires a calibrated camera.
- **Minimum evidence:** an alert is sent only after the person was seen in at least 3 analysed frames.